

BELLSHILL AGCM, AB, CanadaWMO: **710100**Lat: **52.5822N**Lon: **111.4653W**Elev: **721**StdP: **92.96**Time Zone: **-7.00 (NAM)**Period: **07-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-30.3	-27.6	-33.6	0.2	-30.2	-30.9	0.2	-27.5	12.6	-6.6	11.4	-6.1	3.4	270	0.706

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.2	29.3	17.3	27.3	16.4	25.5	15.8	19.2	26.0	18.1	24.9	17.1	23.1	4.8	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.7	13.0	22.3	15.6	12.1	21.0	14.6	11.3	19.9	57.6	26.1	53.9	24.7	50.7	23.2	22.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.1	10.5	9.2	DB	-34.1	32.4	2.6	2.3	-35.9	34.0	-37.4	35.3	-38.8	36.6	-40.7	38.3	(4)
(5)			WB	-34.2	21.3	2.5	1.3	-36.0	22.2	-37.5	23.0	-38.9	23.7	-40.7	24.6	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.3	-10.4	-11.2	-4.4	3.5	11.1	15.0	17.6	16.5	11.7	4.6	-4.2	-11.4	(6)	
	DBStd	12.30	8.53	8.10	7.67	5.35	4.53	3.28	3.06	3.34	4.72	5.13	7.16	7.84	(7)	
	HDD10.0	3226	633	593	447	202	42	4	0	1	35	177	428	664	(8)	
	HDD18.3	5556	891	827	706	444	226	109	51	74	203	424	677	923	(9)	
	CDD10.0	774	0	0	0	8	76	153	236	204	86	11	1	0	(10)	
	CDD18.3	63	0	0	0	0	2	8	29	19	4	0	0	0	(11)	
	CDH23.3	948	0	0	0	3	73	104	366	287	112	3	0	0	(12)	
	CDH26.7	233	0	0	0	0	9	16	108	71	29	0	0	0	(13)	
(14) Wind	WSAvg	4.3	4.5	4.4	4.7	5.1	5.0	4.3	3.6	3.6	4.0	4.3	4.4	4.3	(14)	
(15) Precipitation	PrecAvg	408	21	13	18	30	42	73	71	55	33	18	18	17	(15)	
	PrecMax	547	61	24	43	71	104	110	151	125	83	48	61	30	(16)	
	PrecMin	261	4	2	1	4	10	17	18	6	1	4	3	2	(17)	
	PrecStd	85	13	7	10	16	24	26	30	24	22	10	13	7	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	6.4	6.9	13.8	22.0	28.1	28.5	32.3	30.9	30.1	23.0	13.9	6.8	(19)	
		MCWB	2.7	3.6	7.3	9.6	13.4	16.6	18.4	17.3	16.0	12.4	6.5	3.3	(20)	
	2%	DB	4.6	3.6	10.3	18.8	25.4	26.2	29.7	28.6	26.6	18.2	9.5	3.7	(21)	
		MCWB	2.0	1.2	5.1	8.6	12.5	15.7	18.6	17.1	14.4	10.0	4.7	0.9	(22)	
	5%	DB	2.9	1.9	6.8	15.3	22.9	24.1	27.2	26.6	23.6	15.1	6.4	1.4	(23)	
		MCWB	0.9	0.0	3.3	6.8	11.8	15.1	17.8	16.6	13.4	8.4	2.8	-1.0	(24)	
	10%	DB	1.0	0.2	4.4	12.4	20.3	22.1	25.0	24.5	20.7	12.3	4.2	-1.0	(25)	
		MCWB	-0.8	-1.2	1.7	5.5	10.5	14.2	17.1	16.0	12.2	6.7	1.4	-2.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.2	3.7	7.8	10.7	14.6	18.6	21.2	20.1	16.9	13.0	7.6	3.5	(27)	
		MCDB	6.0	6.6	13.1	21.3	25.3	25.4	27.8	26.3	26.2	22.2	13.1	6.3	(28)	
	2%	WB	2.1	1.4	5.1	8.8	13.3	17.2	19.7	18.7	15.1	10.9	5.1	1.1	(29)	
		MCDB	4.4	3.2	9.4	17.7	22.5	23.3	26.6	25.3	24.6	16.2	8.8	3.5	(30)	
	5%	WB	0.9	0.2	3.3	7.3	12.2	15.9	18.6	17.5	13.9	9.0	3.2	-0.8	(31)	
		MCDB	2.9	1.8	6.6	14.2	21.3	21.8	25.5	24.4	21.9	14.2	6.0	1.1	(32)	
	10%	WB	-0.7	-1.3	1.9	5.9	11.2	14.9	17.6	16.5	12.8	7.3	1.7	-2.5	(33)	
		MCDB	1.0	0.3	4.3	11.8	19.1	20.7	23.8	22.9	20.1	12.0	3.7	-1.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.1	9.1	9.0	10.9	12.7	11.3	12.2	12.5	12.7	10.7	8.6	8.2	(35)	
		MCDBR	9.1	10.0	11.8	16.3	16.3	14.0	15.9	16.4	17.8	15.4	11.1	10.1	(36)	
	5% WB	MCWBR	7.3	8.2	7.6	8.1	6.8	6.2	7.0	7.1	7.6	8.3	7.1	8.0	(37)	
		MCWBR	9.4	9.9	11.1	14.9	14.5	11.9	14.2	14.2	16.0	14.7	10.5	9.7	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.263	0.298	0.292	0.334	0.354	0.368	0.361	0.376	0.318	0.282	0.258	0.243	(40)	
		taud	2.296	2.239	2.420	2.352	2.342	2.340	2.394	2.329	2.457	2.538	2.457	2.341	(41)	
	Ebn at Noon		740	808	909	902	896	883	883	847	867	842	761	710	(42)	
		Edh at Noon	60	87	90	110	117	118	110	112	86	64	51	48	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.09	2.15	3.61	4.75	5.79	5.97	6.30	5.12	3.62	2.16	1.09	0.82	(44)	
	RadStd		0.10	0.16	0.27	0.45	0.39	0.41	0.31	0.42	0.38	0.29	0.11	0.07	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (75 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page